INCH-POUND

MIL-F-63281A(AR)
AMENDMENT 3
16 August 1996
SUPERSEDING
AMENDMENT 2
28 May 1991

MILITARY SPECIFICATION

FUZE, ROCKET: M433 LESS BOOSTER AND BOOSTER LEAD

This Amendment forms a part of Military Specification MIL-F-63281A (AR), dated 14 April 1988, and is approved for use by the U.S. Army Armament Research, Development and Engineering Center, and is available for use by all Departments and Agencies of Defense.

PAGE 2

2.1.1, Add the following after “2-R-376 - 2.75 IN ...”:

“9324106 - Rocket Management Subsystem, Inventory - Deployment, M138”.

PAGE 4

3.3.4, Delete “9332603” and substitute “12551812”.

3.4.1.1, Delete in its entirety.

PAGE 5

3.4.1.2.b, Line 4, before: “The time delay shall be the time interval ...”, add the following: “After the setting charge has been applied, wait 7 sec and then short pins 2 and 4 together.”.

PAGE 6

3.4.2, Delete in its entirety.

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PAGE 7

3.6.1.1, Delete in its entirety.

3.6.1.2, Delete “5 pounds minimum” and substitute “2.5 pounds minimum”.

3.6.2, Delete “9260756” and substitute “12551818”.

3.6.3, Delete “9260756” and substitute “12551818”.

Delete TABLE 11 in its entirety and substitute the following:

“TABLE II

Position of S&A Dev.	Terminals	Requirements
safe	#1 & #4 (charge switch)	100 Milliohms Max
Armed	#1 & #4 (charge switch)	30 Megohms Min @ 500 VDC
Armed	#3 & #5 & Rotor	100 Milliohms Max”.

3.7.1, Delete “speed chip” and substitute “speed clip”.

Delete “#3 and #4” and substitute “#2 and #3”.

PAGE 8

* 3.7.2.2, delete “0.92 and 1.19 seconds” and substitute “1.15 and 1.57 seconds”.

3.8.2, Delete in its entirety and substitute:

“3.8.2 Charge circuit continuity (fuze unarmed). The unarmed fuze shall be tested for continuity through its umbilical lead. The contractor shall design a charge circuit continuity test set that provides an output pulse of 40 mamps maximum for a period of 0.10 msec maximum. The accept/reject requirement of the umbilical cable and the charge circuit of the firing capacitor is:

a. Reject as an open circuit if the measured resistance is greater than the resistance of a 2.25 kohm resistance connected in series with the charge circuit of the firing capacitor.

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b. Reject as a short if the measured resistance is less than the resistance of 220 ohm resistance connected in parallel with the charge circuit of the firing capacitor.

The test set shall be calibrated with acceptable production modules using the modifications below. Thirty fuzes shall be modified for the short condition and thirty fuzes shall be modified for the open condition. The results of the calibration process will be subject to government review and approval."

3.8.3, After "exhibit corresponding function time delays," add "(600 Ms.)".

PAGE 9

3.8.5.3, Add to the end of paragraph "To satisfy the requirement of MIL-STD-331, the disassembled fuze must display the following conditions:

a. No loose explosive material shall be present in the fuze body assembly.

b. The lock roller pin on the rotor shall not be broken.

c. The rotor shall be in the safe position, secured such that it cannot be turned to the armed position with finger pressure.

d. The g-weight shall be in the fully safe position.

PAGE 10

3.8.5.5, Add to the end of paragraph "To satisfy the safety requirement, the disassembled fuze must display the following conditions:

a. No loose explosive material shall be present in the fuze body assembly.

b. The lock roller pin on the rotor shall not be broken.

c. The rotor shall be in the safe position, secured such that it cannot be turned to the armed position with finger pressure.

d. The g-weight shall be in the fully safe position.

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e. The Ultem insulator switch post shall not be completely separated from the setback weight assembly."

PAGE 12

4.3.1, Delete "9332604" and substitute "12551818".
Add ", Detonator" after Contact and before 9332598.
Add ", Ground" after Contact and before 9332599.

Under Fuze Rocket Quantity, Delete "366" and substitute "246".

* Under Part Description, Molded Electronics Assembly (Molded), delete "702" and substitute "164".

PAGE 15

Table III: Delete second "Gear and pinion assembly No.1" and substitute "Gear and pinion assembly No. 2".

PAGE 18

Table III: Delete "9260752" and substitute "12551817".
Delete "Pull-off test of switch cap 25 3.6.1.1 4.5.32".

PAGE 19

Table III: Delete "9332604" and substitute "12551818".

* Table III, under Molded Electronics Assembly,
Thermal shock, delete "72" and substitute "24".
Module turn on time, delete in its entirety.
Module time delay function, delete "702 4.5.31" and substitute "164 4.5.34".

PAGE 20

Table III: Under Fuze, Rocket M433 Less Booster and Booster Lead, No. of sample units, delete "366" and substitute "246" in two places.

PAGE 21

Table III: Under Five Foot Drop Test, delete "25" and substitute "24".

Under Non-environmental test, delete "67" and substitute "7".

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Under Aircraft vibration, helicopter, delete “81” and substitute “63”.

Under Temperature and humidity test, delete “21” and substitute “22”.

Under Ballistic firing test, delete “138” and substitute “120”.

Under Transportation vibration, delete “120” and substitute “78”.

PAGE 25

4.4.2.1, Add Majors 118, 119 and 120 as follows:

“118	Wrong lubricant	100%	3.3	Certification	
119	Excessive lubricant (a)	0.40%	3.3	Visual	120
Lubricant in not-called-for area		0.40%	3.3	Visual”.	

Add note to note block as follows:

“(a) Excessive lubricant is defined as evidence of non-uniform coating and collection of fluid in visible areas.”

PAGE 26

* 4.4.2.1, delete Major 111, “Marking missing or incomplete...” in its entirety.

PAGE 28

* 4.4.2.2, delete Minor 201, “Marking missing or incomplete...” in its entirety.

PAGE 29

4.4.2.3, Delete “9260750” and substitute “9260751”.

PAGE 31

4.4.2.5, Delete “9260774” and substitute “9260751”.

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PAGE 37

4..4.2.11, Delete “9260751” and substitute “9260774”.

PAGE 38

4.4.2.12, Delete “and Gear”.

PAGE 48

* 4.4.2.21, delete Minor 111, “Module turn on time...” in its entirety.

PAGE 50

4.2.22, Add Critical 2 as follows:

“2	Pull-off test of switch assembly (a)	100%	3.6.1.2	4.5.33”

Delete Majors 102 and 103 in their entirety.

Delete note a in its entirety and substitute:

“(a) Perform Critical 2 before critical 1.”

PAGE 51

4.4.2.23, Delete “9332604” and substitute “12551818”.

Major 102, Delete “#1 and #5” and substitute “#1 and #4”

Delete Major 103 in its entirety.

Major 105, Delete “#1 and #5”, and substitute “#1 and #4”.

PAGE 52

4.4.2.24, Delete “9332604” and substitute “12551818”.

Add majors 108 and 109:

“108	Width of setback weight actuator, pin clearance (.110-.005)	0.40%	3.3	Gage

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109	Height of setback weight actuator pin clearance (.420+.010)		0.40%	3.3	Gage”.

PAGE 53

4.4.2.25, Delete “9332603” and substitute “12551812”.

Delete “9332604”, and substitute “12551818”.

Delete Majors 102, 103, 104 and substitute Major 102 as follows:

“102 Alignment of contacts 0.40% 3.3 Visual”.

Delete notes in their entirety.

PAGE 56

4.4.2.28, Add Majors 104, 105, 106, 107 and 108 as follows:

“104	Wrong adhesive	100%	3.3	Certification
105	Excessive adhesive (a)	0.40%	3.3	Visual
106	Adhesive in not-called-for area	0.40%	3.3	Visual
107	Improper cure (time & condition) of adhesive	0.40%	3.3	Visual/chart
108	Improper bonding of contact	0.40%	3.3	Visual”.

Delete Minor 201 in its entirety.

Add note to notes block as follows:

“(a) Adhesive is permitted in the immediate area around the contact/module interface as long as it does not extend into the contact cavity, make contact with the solder, or cause interference fit with the next assembly.”

PAGE 62

4.4.2.32, Delete “9254707” and substitute “9260886”.

Delete Major 101 in its entirety.

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PAGE 63

4.4.2.33, Under Critical 7, Non-environmental test, delete "38" and substitute "8".

PAGE 64

4.4.2.33, Under Major 104, Ballistic fuze firing set, delete "90" and substitute "60".

Add Major 108 as follows:

"108 Thermal shock 20 3.8.4.1 4.5.29".

Delete note c in its entirety.

* Under Major 106, Charge Circuit Continuity, delete "54" and substitute "63".

* Under Major 107, delete "120" and substitute "30".

PAGE 65

4.4.2.34, Add Majors 109, 110, 111, 112, 113, 114, 115 and 116 as follows:

"109	Wrong umbilical adhesive		100%	3.3	Cert.
110	Umbilical adhesive missing	0.40%	3.3	Visual	
111	Umbilical adhesive excessive (a)		0.40%	3.3	Visual
112	Umbilical adhesive in not-called-for area	0.40%	3.3	Visual	
113	Wrong impact probe sealant	100%		3.3	Cert.
114	Impact probe sealant missing	0.40%	3.3	Visual	
115	Impact probe sealant excessive (b)		0.40%	3.3	Visual
116	Impact probe sealant in not-called-for area		0.40%	3.3	Visual".

Delete Minors 201 and 202 in their entirety.

Add notes to note block as follows:

"(a) Evidence of adhesive inside cavity of fuze body or failure to meet note 6 shall constitute excessive adhesive.

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(b) Sealant is permitted in the immediate area around the joint of the probe and fuze body as long as it does not extend radically inward more than 0.10 inch or above the crimp lip."

PAGE 68

4.4.2.37, Delete "9332603" and substitute "12551812".

Delete Major 101 in its entirety.

PAGE 69

4.4.2.38, Delete "9332603" and substitute "12551812".

Delete title "Contact" and substitute "Contact, Detonator".

PAGE 70

4.4.2.39, Delete "9332603" and substitute "12551812".

Delete title "Contact" and substitute "Contact, Ground".

PAGE 79

4.5.16, 7th line delete "three (3) seconds" and substitute "five (5) seconds".

PAGE 84

Add "4.5.24.1 Correlation procedure." before "For the correlation procedure,...".

PAGE 86

* 4.5.30, Delete in its entirety.

4.5.32, Delete in its entirety.

4.5.33, Delete in its entirety and substitute:

"4.5.33 Pull-off test of switch assembly, Secure the setback weight in a holding fixture. An axial load of 2.5 pounds minimum shall be applied to the switch cap. The total height of the pin, insulator and cap assembly shall not increase by more than .001 in. due to this pull test. Any unit that fails this requirement shall be classified as defective."

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PAGE 87

4.5.35, Line 5, delete “MK40, Mod 3” and substitute “MK66, all MOD.” .

4.5.35.3, Delete “200 and 460 msec settings” and substitute “450 msec setting”.

PAGE 88

4.5.35.5, Delete 200 msec nominal time setting in its entirety.

Under 460 msec nominal time setting, delete “460 msec” and substitute “450 msec”; delete “50 to 1075 feet” and substitute “28 to 1088 feet”.

4.3.35.6.1, Delete in its entirety and substitute the following:

“4.3.35.6.1 First article sample. If three (3) or more units do not meet the requirements in the superquick and the 450 msec canopy mode, or five (5) or more units do not meet the requirements in the 20 msec bunker mode of the operating mode tests of 4.5.35.5, or two (2) or more units fail the charge circuit continuity test, or eight (8) or more units fail the overall ballistic test (Table VI), the first article ballistic sample shall be rejected.

4.3.35.6.2, Delete in its entirety and substitute the following:

“4.3.35.6.2 Lot acceptance sample. If two (2) or more units do not meet the requirements in the superquick and the 450 msec canopy mode, or three (3) or more units do not meet the requirements in the 20 msec bunker mode of the operating mode tests of 4.5.35.5, or two (2) or more units fail the charge circuit continuity test, or five (5) or more units fail the overall ballistic test (Table V11), the lot acceptance ballistic sample shall be rejected.”

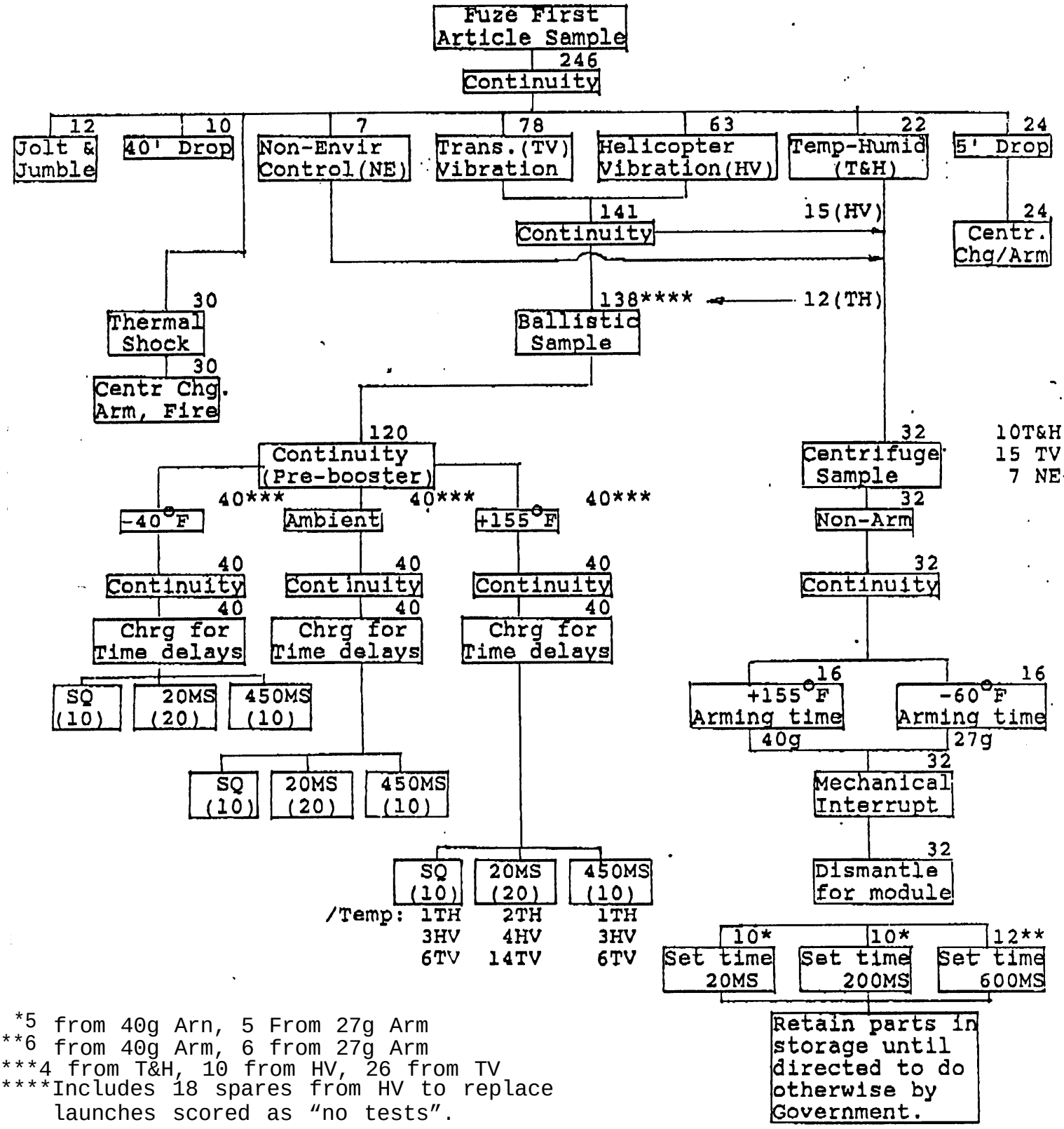
PAGE 89

5.2, Delete “Dwg. 8886458” and substitute “Dwg. 8861213”.

PAGE 91

Delete PAGE 91 in its entirety and substitute enclosed Page 11 (91).

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*5 from 40g Arn, 5 From 27g Arm
**6 from 40g Arm, 6 from 27g Arm
***4 from T&H, 10 from HV, 26 from TV
****Includes 18 spares from HV to replace launches scored as “no tests”.

Note: A setter that represents the helicopter setter system shall be used for delay time acceptance tests in all cases.

TABLE VI First article test.

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PAGE 92

Table VI: Delete paragraph 3 in its entirety.

Delete paragraph 4 in its entirety.

* Delete the environment and module quantity and substitute with notes as follows:

Nominal Time Settings	Environment	Module Quantity	Acceptance Limits
20ms	-60 F	70 *	5 - 35 msec
	ambient & +155 F	70 **	
	Thermal shock	24 ***	
200ms	-60 F	70 *	185-215msec
	ambient & +155 F	70 **	
	Thermal shock	24 ***	
600ms	-60 F	70 *	570-630msec
	ambient & +155 F	70 **	
	Thermal shock	24 ***	

* All 70 units go through the -60 F environment at all three nominal time settings.
** All 70 units go through the ambient environment at all three nominal time settings and then go through the same test at +155 F.
*** All 24 units go through the thermal shock at all three nominal time settings.

NOTE: The firing capacitor shall be checked to insure that at least minimum detonator firing output pulse is present when the detonator firing circuit is tripped at the conclusion of the time delay.

PAGE 93

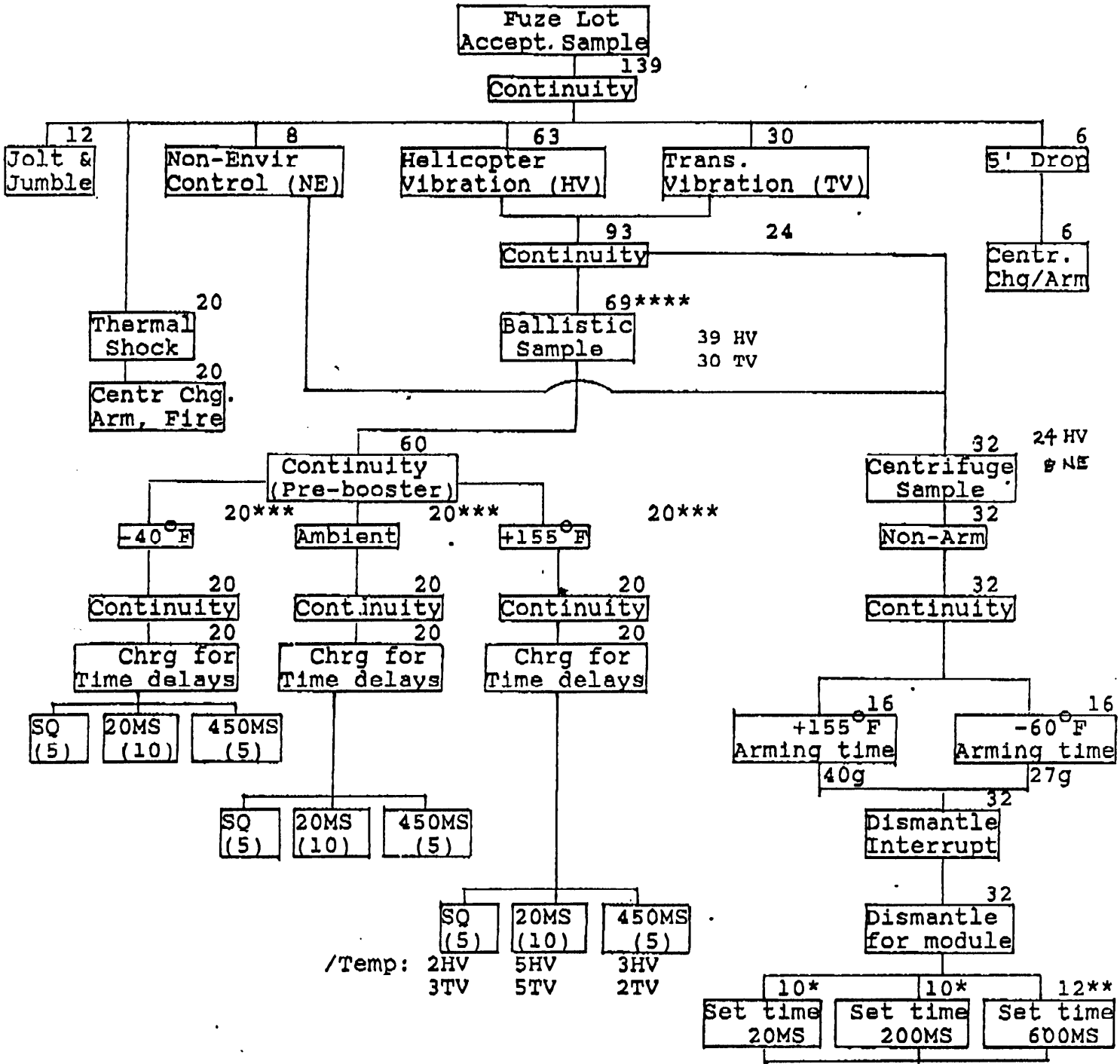
Table VI: Delete paragraph 6 in its entirety.

Paragraph 8, delete “Both setters shall be used.”

PAGE 94

Delete Table VII in its entirety and substitute enclosed Page 13 (94).

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* 5 from 40g Arm, 5 from 27g Arm
**6 from 40g Arm, 5 from 27g Arm
***10 from HV, 10 from TV
**** Includes 9 spares from HV to replace launches scored ae no-tests.

Note: A setter that represents the helicopter setter for delay time acceptance tests in all cases.

TABLE VII Fuze lot acceptance test.

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